Nibraas Khan

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Highlights

Shipping real-time ML from lab to product (time-series, vision, and LLMs) delivering <1s inference, 80%+ F1, and AWS-scale MLOps.

Education

- **Vanderbilt University** — *Aug 2020 - Dec 2025 Expected*
  Ph.D. Computer Science — Nashville, TN
- **Middle Tennessee State University** — *Aug 2017 - May 2020*
  B.S. Computer Science — Murfreesboro, TN

Skills

- **Languages:** Python, Typescript, C#, Kotlin, SQL, Bash
- **ML Frameworks & Data Science:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Hugging Face Transformers, OpenCV
- **Cloud & MLOps:** AWS (EC2, S3, SageMaker), Firebase, GCP, Docker, Git, GitHub Actions, Git LFS
- **Data Engineering & Orchestration:** Kafka, Airflow, Great Expectations, N8N
- **Databases:** MongoDB, SQLite
- **ML Specializations:** Explainable AI (xAI), Reinforcement Learning (Hierarchical RL, Policy Transfer), Deep Learning (LSTMs, GANs, Siamese Networks), Computer Vision (Object Detection & Tracking), Time-Series Analysis (Wearable Sensor Data), LLM Fine-Tuning, RAG, embeddings, prompt engineering
- **Developer Platforms & Tools:** Unity, React, React Native, Android/iOS Development, Core ML, TensorFlow Lite, Flask, WebSockets, Pytest

Experience

- **Vanderbilt University - Robotics and Autonomous Systems Lab** — *May 2020 - Present*
  Machine Learning Engineer (Research) — Nashville, TN
  - **Tiered eXplainable AI (xAI) Framework**
    - Shipping a Python xAI library that discovers motion primitives from time-series (kernel CPD and HDBSCAN), aligns them to prototypes, and produces tiered explanations for stakeholder review.
    - Automating developer and stakeholder views with saliency over raw signals, primitives visualizations, prototypes, and counterfactual exploration for model audits.
    - Providing model-agnostic APIs that integrate with PyTorch and TensorFlow and fit standard MLOps workflows; delivering clear documentation and examples.
  - **Real-Time Precursor Detection System (REACT)**
    - Training few-shot PyTorch model in low-data clinical environment; engineering a multimodal feature pipeline from wearable time-series and audio into 50+ features; tuning with Optuna to reach 84.6% F1.
    - Operating MLOps stack on AWS EC2 with Kafka ingestion; using Great Expectations for data quality; orchestrating weekly retraining in Airflow trains in Docker and deploys artifacts for to an iOS fleet.
    - Shipping an iOS app for data collection, real-time inference, and alerts (Apple Watch); using xAI library to surface prototype matches and counterfactuals; logging outcomes for continuous model monitoring.
  - **End-to-End Wearable System for Biomechanical Analysis**
    - Building the full stack from hardware to firmware to cloud to app; developing a custom PCB with IMUs/ESP32, streaming firmware, ingestion and services, and a React app with a 3D avatar.
    - Leading data collection and real-time analysis on AWS; streaming 100 Hz from 10 IMUs to the react app and AWS; delivering analysis to feedback in <1s for athlete guidance.
    - Using the xAI library to explain movements as motion primitives; comparing against expert movement with a Siamese model; driving actionable feedback for training sessions for any sport or activity.
  - **Deep Learning Pipeline for Digital Biomarker Discovery**
    - Training a TensorFlow/Keras Bi-GRU with 4-head attention to model activity sequences and identify biomarkers differentiating MCI vs healthy for clinical insights; tuning with Optuna; reporting 86% F1.

- ∗ Architecting pipelines for time-series analysis; leading a coding team for data labeling; fixing unsynced IMU starts, misaligned timestamps, and missing labels; delivering reproducible datasets and code.
- ∗ Applying the xAI library to decompose sequences into motion primitives and generate explanations.

◦ **Mentorship & Guidance**

- ∗ Scaled a cross-institution mentorship program; mentored 30+ students (2 to PhD, 3 to master's); built onboarding docs, starter repos, and tests that improved reproducibility and delivery velocity.
- ∗ Led ownership for teams from universities building research components; delivered components of an ASD driving simulator and VR therapeutic activities; ran agile sprints, code reviews, and specs.
- ∗ Mentored Vanderbilt undergraduates and peers applying machine learning to estimate biomechanical parameters; taught prompt engineering; guided hardware, firmware, and mobile-app development.

- **MTSU's Phillip Lab** *Aug 2018 – May 2020*
  Research Assistant Murfreesboro, TN
  - ◦ Developed a hierarchical reinforcement-learning framework fusing cue-guided working memory and feedback-based task switching for non-Markovian tasks; enabled policy transfer.
  - ◦ Implemented a task-representation switch with temporal-difference learning using positive/negative prediction errors; encoded state, cues, working memory with holographic reduced representations.
  - ◦ Improved accuracy on combined tasks from 38.33% to 96.88% through hyperparameter tuning with Bayesian optimization and rigorous evaluation.

VENTURE CREATION

- **JumpStart** *Jan 2023 – Present*
  Co-Founder Nashville, TN
  - ◦ Co-founded a community program where students gain career-ready skills by building pro-bono software for real clients, supported by technical workshops and industry mentorship.
  - ◦ Guided students to ship a real-time UAV ground-control station; integrated YOLOv8 computer-vision pipeline with DeepSORT tracking and low-latency WebRTC streaming.
  - ◦ Co-mentored students to develop and deploy an XGBoost clinical analysis tool for Vanderbilt's TRIAD; adding online learning that continuously improves as new data arrive.

SELECTED PUBLICATIONS

- **Khan, N.**, Wang, D., Ghosh, R., Tauseef, M., Mion, L., Ma, M., & Sarkar, N. (2025). Decoding Human Motion: A Scoping Review of Explainable AI Methods in Movement Analysis. Pervasive Computing Technologies for Healthcare. In-Review.
- **Khan, N.**, Shragge, I., Zilinskaite, G., Plunk, A., Staubitz, J., Rajaraman, A., Weitlauf, A., & Sarkar, N. (2025). Interpretable Deep Few-Shot Learning for Prediction of Precursors to Challenging Behaviors in Individuals with Intellectual and Developmental Disabilities. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies. In-Review.
- **Khan, N.**, Plunk, A., Zheng, Z., Adiani, D., Staubitz, J., Weitlauf, A., & Sarkar, N. (2024). Pilot study of a real-time early agitation capture technology (REACT) for children with intellectual and developmental disabilities. Digital Health, 10. https://doi.org/10.1177/20552076241287884
- Wang, H. D., **Khan, N.**, Chen, A., Sarkar, N., Wisniewski, P., & Ma, M. (2024). MicroXercise: A Micro-Level Comparative and Explainable System for Remote Physical Therapy. IEEE. https://doi.org/10.1109/chase60773.2024.00017
- **Khan, N.**, Tauseef, M., Ghosh, R., & Sarkar, N. (2024). A Novel Loss Function Utilizing Wasserstein Distance to Reduce Subject-Dependent Noise for Generalizable Models in Affective Computing. International Conference on Human-Computer Interaction. https://doi.org/10.1007/978-3-031-61572-6_2

ADDITIONAL INFORMATION

Languages: Fluent in English, Conversational in Hindi and Urdu
Interests: Cooking, Boxing, Exploring restaurants